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Chemical foundation model-guided design of high ionic conductivity electrolyte formulations



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Designing optimal formulations is a major challenge in developing electrolytes for the next generation of rechargeable batteries due to the vast combinatorial design space and complex interplay between multiple constituents. Machine learning (ML) offers a powerful tool to uncover underlying chemical design rules and accelerate the process of formulation discovery. In this work, we present an approach to design new formulations that can achieve target performance, using a generalizable chemical foundation model. The chemical foundation model is fine-tuned on an experimental dataset of 13,666 ionic conductivity values curated from the lithium-ion battery literature. The fine-tuned model is used to discover 7 novel high conductivity electrolyte formulations through generative screening, improving the conductivity of LiFSI- and LiDFOB-based electrolytes by 82% and 172%, respectively. These findings highlight a generalizable workflow that is highly adaptable to the discovery of chemical mixtures with tailored properties to address challenges in energy storage and beyond.

Discovering novel chemical mixtures with tailored properties is a critical roadblock to the development of new technologies in drug discovery, environmental remediation, catalysis, cosmetics, and electrolytes for rechargeable batteries^{1–5}. The formulation design space grows exponentially with each additional component, and interactions between components result in a complex mapping of their chemical and compositional design to a target property^{6,7}. As a result, formulation design and optimization are often time-consuming trial-and-error processes.

In the case of battery electrolyte formulations, where research can incur significant materials and testing costs, methods that reduce the number of experiments to reach optimal designs would be extremely valuable⁸. Currently, LiPF₆ is the industry standard salt due to its balance of solubility, transport properties, stable solid electrolyte interphase (SEI) formation ability, and chemical stability^{9,10}. Trial-and-error testing over the last 30 years has optimized the mixture of linear and nonlinear carbonate solvents to pair with LiPF₆ in battery electrolytes^{11,12}. Several promising alternative salts, such as LiDFOB, offer advantages in terms of thermal stability and SEI stabilization, but suffer from low conductivity in carbonate-based solvents¹³. Insufficient conductivity has been identified to contribute to degradation in commercial battery chemistries via lithium metal plating on the graphite anode^{12,14–16}. Therefore, identifying solvent mixtures that enhance ionic conductivity is a critical step toward the adoption of new salts. Improving ionic conductivity remains a challenging problem in electrolyte research due

to its complex relationship with the molecular geometry of individual constituents in a formulation^{17–19}. Addressing this challenge requires efficient searching of the vast and mostly unexplored formulation design space.

The emergence of machine learning (ML) has ushered in a new paradigm of models that leverage data-driven approaches to learn structure-property relationships for the design of novel battery materials^{20–22}. Fewer studies have focused on extending these approaches to electrolyte formulations due to the indirect relationship between formulation property and the properties of constituent molecules^{23–31}. One straightforward approach to optimizing electrolyte concentration is to use a surrogate model like a Gaussian process regression model that can guide the exploration of a fixed component system based on the concentration of components^{29–31}. These black-box methods have often been paired with high-throughput platforms to guide optimization campaigns faster than design of experiment (DOE) and random searching, but are limited to interpolation within a design space of fixed components. To continue design exploration outside fixed components, using chemical motifs or molecular properties as descriptors is a more favorable approach. However, defining and compiling such descriptors could be difficult in the vast chemical space of battery-relevant molecules^{32–34}. This issue can be overcome by using models that can directly map the molecular structure to the target property, thus basing the ML input on the underlying physical chemistry, making the approach generalizable to the vast chemical design space beyond the limitations of available data or descriptors^{35,36}. An

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example would be graph convolution networks (GCNs) that can operate on molecular graphs to encode molecular topology as an input. This approach has been used to predict properties of molecules, materials, mixtures^{23–26}, as well as battery performance^{27,28}. To scale the GCN for formulations, the graph representations of the molecules in each formulation are aggregated into a formulation descriptor that is updated during the training process. We have previously demonstrated the advantages of pre-training a GCN on molecular properties, such as HOMO–LUMO energy levels and electric moment, on the accuracy of predicting formulation performance in small-data regimes common for experimental battery data^{27,28}. Though the use of pre-trained molecular structural descriptors (like GCN) in a formulation representation overcomes the issue of high degrees of freedom and data limitations, this approach is not efficient due to the reliance on the labeled datasets (HOMO–LUMO and electric moment for molecules) for the pre-training step. Additionally, the predictive accuracy of these models relies on the relevance of pre-training labels toward the formulation property of interest, therefore limiting the generalization of the approach. In recent years, transformer-based models trained on large unlabeled text-based representations of molecules (SMILES) have been transformative for the cheminformatics field due to their success in predicting a wide range of molecular properties with high accuracy^{37–40}. By utilizing self-supervised learning capabilities, these chemical foundation models reduce the dependence on labeled data and task-specific features⁴¹. Despite this, the customization and use of these chemical language models for the development of advanced material systems, such as formulations, remains limited^{42,43}.

In this paper, we introduce a data-driven model to predict the ionic conductivity (IC) of liquid electrolyte formulations, called **SMI-TED-IC**, that is based on a previously published and open-sourced chemical foundation model for small molecules, SMI-TED (SMILES Transformer Encoder Decoder). We customize a string representation of formulations and fine-tune SMI-TED⁴⁴ with a curated dataset of 13,666 electrolyte formulations and their respective ionic conductivity measurements from the lithium-ion battery literature^{45–51}. The model is then used to screen through

10^5 computationally generated electrolytes and identify novel electrolyte formulation designs with improved ionic conductivity that are further validated experimentally. The chemical language model-aided formulation discovery workflow is schematically summarized in Fig. 1a. The generalizability of foundation models across chemical space enables the exploration of new formulation designs with optimal properties.

Results

Analysis of literature curated dataset

To capture the complex relationship between electrolyte composition and ionic conductivity, we curate a chemically diverse dataset of 13,666 electrolyte formulations and their reported ionic conductivities at different temperatures from the lithium-ion battery literature (Table S1). These electrolyte formulations are solutions of charge-carrying lithium salts and aprotic organic solvents that have been investigated for use in various lithium-based chemistries. There are 15 unique salts in the dataset, with LiPF₆, LiBOB, and LiBF₄ being the most prominent ones (Fig. S1a). At least one salt is present in every electrolyte, and the total salt concentration ranges between 5 and 15 mol% (Fig. S1b). The optimal solvent-salt pairings are dependent upon several physiochemical aspects such as solubility, chemical compatibility, and ion-solvent interactions in the solvation layer⁵². Although it is hard to establish a quantitative trend between these properties and the chemical structure of salts and solvents due to complex interplaying relationships, heuristic design rules based on chemical functional groups have been useful in identifying synergistic combinations of salts and solvent, such as the common combination of LiPF₆ with carbonate-based solvents⁷. The dataset includes 51 unique solvents with diverse chemical structures to provide a broad spectrum of the chemical design space of lithium-ion battery-relevant solvents. For analysis, solvents are grouped into 12 solvent classes based on their chemical similarity and labeled by the primary functional group present in the compound (i.e., carbonates, ethers, etc.). Figure 2a shows a mapping of the chemical structure of the solvents using MACCS keys representations, projected into two dimensions using

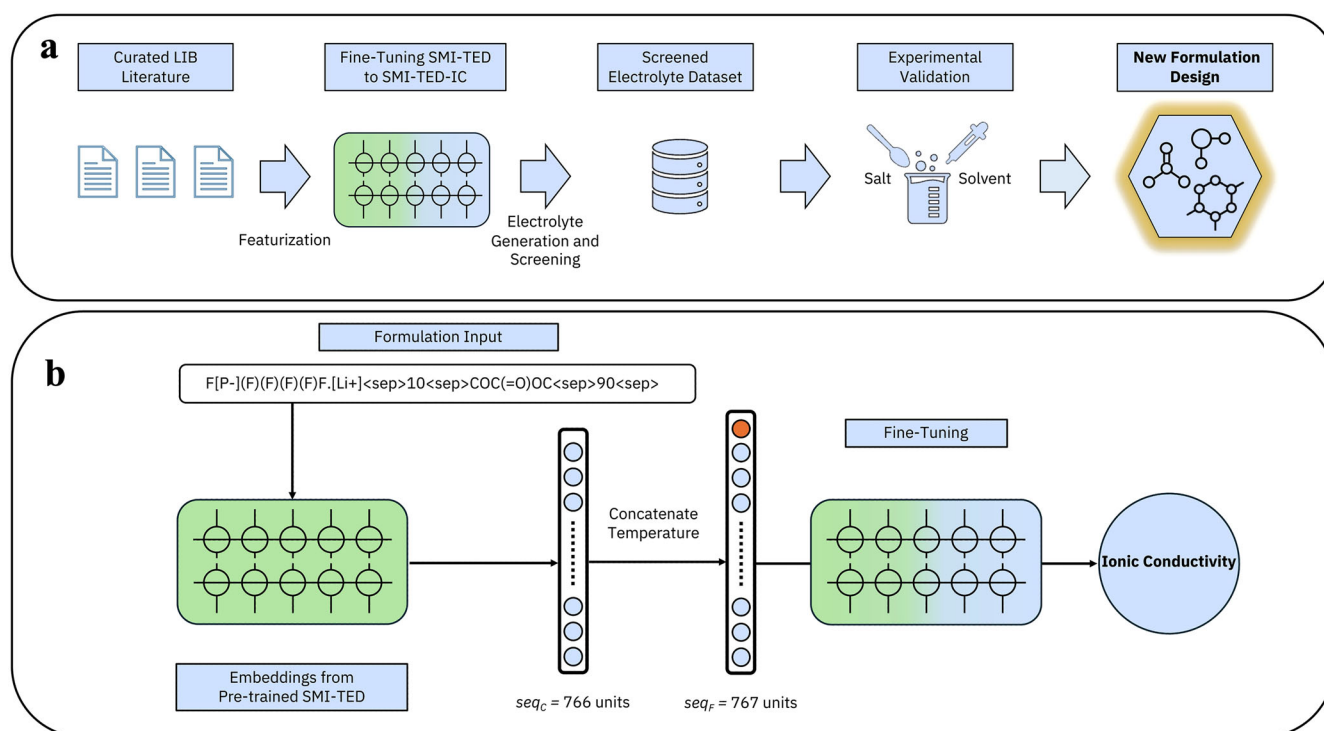


Fig. 1 | Schematic illustration of formulation design workflow and input featurization. a Chemical foundation model (SMI-TED) guided discovery workflow for electrolyte formulations. **b** Schematic describing fine-tuning of the SMI-TED model to ionic conductivity predicting SMI-TED-IC. The string input for SMI-TED

is adapted for formulation design that includes both SMILES and constituent composition. The model is fine-tuned to predict ionic conductivity after concatenation with the temperature.

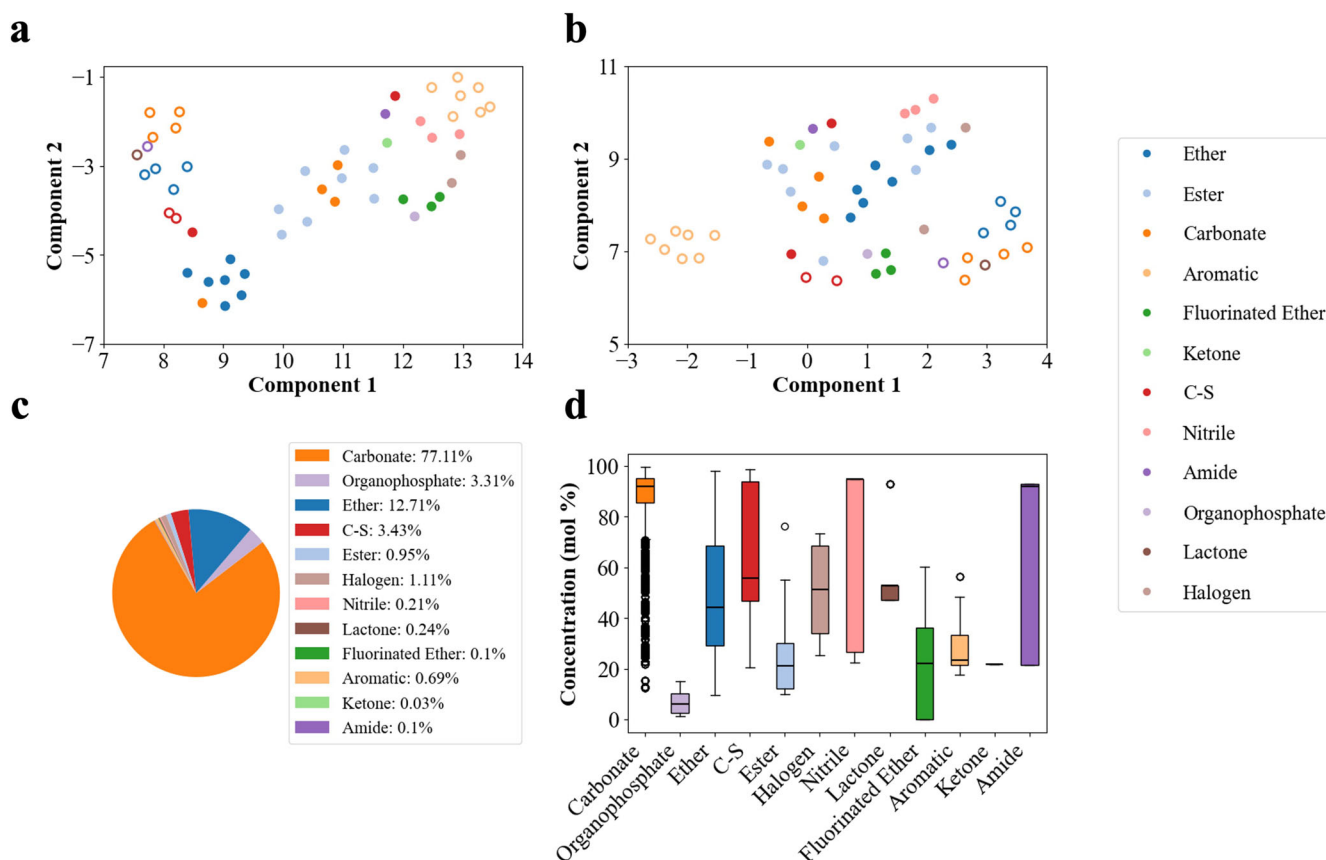


Fig. 2 | Visualization of solvent molecule design space and composition in the dataset. Two dimensional (2D) UMAP projections of **a** MACCS fingerprints of the solvents in the dataset and **b** SMI-TED embeddings of the solvents in the dataset. The solvents are colored by the assigned solvent classes with open circles corresponding

to cyclic compounds and solid circles corresponding to linear compounds.

c Percentage of electrolyte formulations that contain an instance of a solvent class.

d The concentration ranges of each solvent class in the dataset.

UMAP⁵³. MACCS keys are composed of 166 chemical substructure keys and are a common method for calculating similarity metrics between molecules. Thus, the proximity of one molecule to another in the UMAP plot suggests similarity in chemical structures⁵⁴. The cyclic compounds (indicated by open circles in Fig. 2a, b) are differentiated from their linear counterparts (solid circles in Fig. 2a, b). It is apparent from the UMAP plots that the solvents within a solvent class are clustered together, although cyclic compounds (open circles) are structurally distinct from their linear counterparts for all solvent classes. This clustering is also captured in the UMAP projection of the SMI-TED embeddings of the solvent molecules (Fig. 2b), which yields a distribution of solvents on the basis of structural information that is embedded in the pre-trained foundation model. The comparison of the two UMAP projections highlights the ability of SMI-TED embeddings to differentiate molecules based on their chemical and structural characteristics. Furthermore, SMI-TED embeddings demonstrate more continuity in the chemical map as evidenced by the convergence of cyclic molecules toward the bottom of the plot in Fig. 2b. This is in contrast to the UMAP projection of MACCS keys in Fig. 2a, where continuity in design space is broken due to the limitations of MACCS keys in being solely focused on individual chemical groups rather than the global chemical nature of the molecule. Chemical representations such as the SMI-TED embedding have been demonstrated to have advantages over fingerprints in encoding the influence of semi-local and global molecular structures, which will influence many downstream prediction tasks⁵⁵.

The distribution of solvent classes in the dataset is shown in Fig. 2c. The large majority (77%) of electrolytes include carbonates due to their prevalence in commercial lithium-ion batteries. The limited representation of other solvent classes in the literature dataset does not significantly diminish

the scope of the present model due to the large-scale pre-training of SMI-TED on 91 million SMILES samples from PubChem. The wide coverage of the chemical information in the pre-training enhances the model's generalization potential to the out-of-distribution formulation designs, as demonstrated in the next sections. The concentration of the constituents is also an important determining factor in the ionic conductivity of the formulation. The optimal solvent concentration in a formulation varies by solvent class (Fig. 2d). For example, the average concentration of carbonates is high (>80 mol%) because they are used as the primary solvent in most electrolytes. In contrast, solvents such as esters and fluorinated ethers are mainly used as lower concentration co-solvents. Solubility limitation is an important factor in these examples, as the solubility of most salts is limited in solvents like fluorinated ethers, and therefore their concentration must be reduced to avoid phase separation. The analysis of chemical diversity in the dataset allows us to establish the learned design space of the model.

Model performance and benchmarking

SMI-TED, a chemical foundation model pre-trained with string representation of 91 million molecules, is fine-tuned with the literature-curated electrolyte formulation dataset. The dataset consists of formulations containing up to six electrolyte constituents, where each constituent is represented by its canonical SMILES string. The concentration of each constituent molecule is expressed as the fraction of its mol% in the formulation. The SMILES of molecular constituents in each formulation are combined with their respective compositions into a string representation as demonstrated in Fig. 1b, where each component is separated by a <sep> token. This string representation of the formulation and the additional feature temperature is used as an input for SMI-TED fine-tuning (referred to

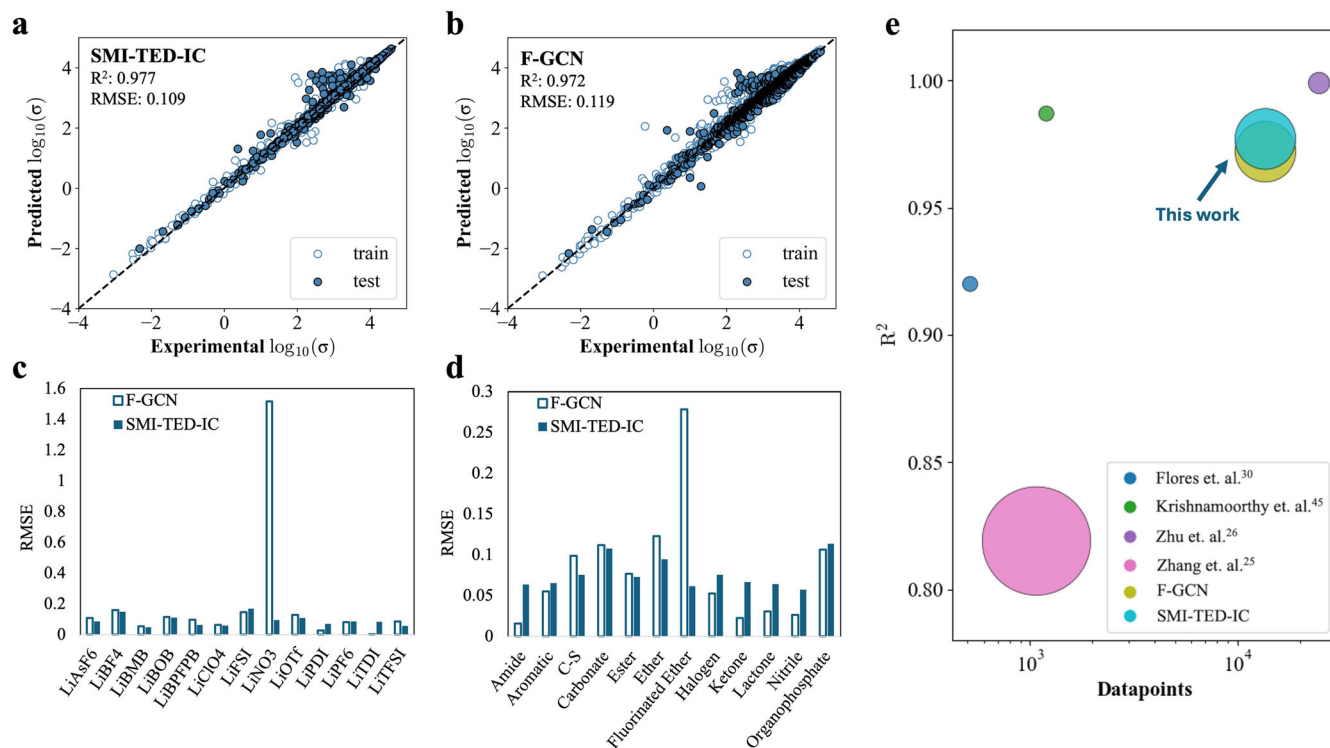


Fig. 3 | Model error benchmarking and comparison to literature. Parity plots showing predicted ionic conductivity vs. the experimental values in the test dataset for **a** the SMI-TED-IC model and **b** F-GCN model. RMSE of both models on the test dataset for electrolytes containing each unique **c** solvent class and **d** lithium salt.

e Comparison of model performance with machine learning models reported in literature for predicting ionic conductivity. The size of the marker corresponds to the chemical diversity in the dataset.

as SMI-TED-IC post fine-tuning) while ionic conductivity is used as an outcome label. The final sequence captures the complete chemical, compositional, and conditional (temperature) information of the electrolyte. Thus, the formulation model SMI-TED-IC is built on the SMI-TED architecture by introducing a formulation-aware input representation and fine-tuning strategy tailored for multicomponent electrolyte mixtures, extending its application to multicomponent formulation properties such as ionic conductivity. This formulation-aware representation allows the model to process both molecular identity and relative composition within a unified input structure, leveraging the same tokenizer and embedding layers used for SMILES sequences. Interleaving composition values with molecular tokens enables joint attention over the full formulation context. Pre-training on millions of SMILES strings allows SMI-TED to learn a chemically-informed latent space that captures molecular substructures and their contextual relationships³⁶. This latent representation provides a strong foundation for fine-tuning on electrolyte formulations, enabling the SMI-TED-IC model to associate moiety-level composition patterns with macroscopic properties such as ionic conductivity. The complete details of data featurization, augmentation, and regression fine-tuning are described in the “Methods” section.

The prediction accuracy of the SMI-TED-IC model is shown in Fig. 3a as a parity plot. The model is tested on a random 10% subset of the complete dataset. The root mean squared error (RMSE) in ionic conductivity prediction for the test data is $0.1087 \log_{10}(\sigma)$. The model performance is benchmarked against a formulation graph convolution network model (F-GCN) using the same training and test data splits. The F-GCN model uses GCNs pre-trained with HOMO-LUMO and electric moment to represent the chemical structure of formulation constituents^{27,28}. In the present analysis, the SMI-TED-IC model outperforms the F-GCN (Fig. 3b) model in conductivity prediction. These results are in agreement with previous reports comparing electrolyte property prediction using chemical

foundation models to other machine learning methods^{42,43}. Still, the accuracy of the F-GCN model is only slightly lower than the SMI-TED-IC model with a test RMSE of $0.1190 \log_{10}(\sigma)$. Both SMI-TED-IC and F-GCN map the underlying structural features that influence molecular properties and, therefore, can be used to predict properties in domains with sparse data. To further probe the capability of each model for this task, we evaluate the error on specific solvent classes and salts (Fig. 3c, d). Generally, the error among solvent classes and salts is similar for both models, except for fluorinated ether and LiNO₃ containing electrolytes, where F-GCN has high prediction errors. These molecules are sparsely represented in the ionic conductivity dataset, with only one example constituting LiNO₃ in the training set. This finding highlights the crucial difference in the scope of F-GCN and the foundation model, which can be attributed to the differences in their pre-training process. As discussed in the “Methods” section, the SMI-TED model was pre-trained with over 91 million molecules using a masked language model method to correctly predict masked molecular tokens based on the rest of the string^{38,57}. The success of these chemical foundation models is attributed to the underlying chemical structure rules that the model learns during this self-supervised training process. This approach equips SMI-TED-IC with a broader understanding of chemical diversity, enabling it to generalize more effectively to previously unseen species, even when training data is sparse or structurally distinct from well-represented regions. This is supported by a comparative experiment in which a transformer encoder was trained to directly map a formulation string and temperature to the ionic conductivity values, skipping the pre-training step. The analysis performed using the same distribution of train, validation and test dataset as SMI-TED-IC shows a higher RMSE value (0.4341 in Fig. S2) than the pre-trained SMI-TED-IC (0.1087). This conclusion is also supported by a recent study on extrapolation in molecular property prediction, which has demonstrated the advantage of self-supervised pre-training to uncover the relative trends of molecular properties⁵⁸. The F-GCN model, on the other hand, incorporates

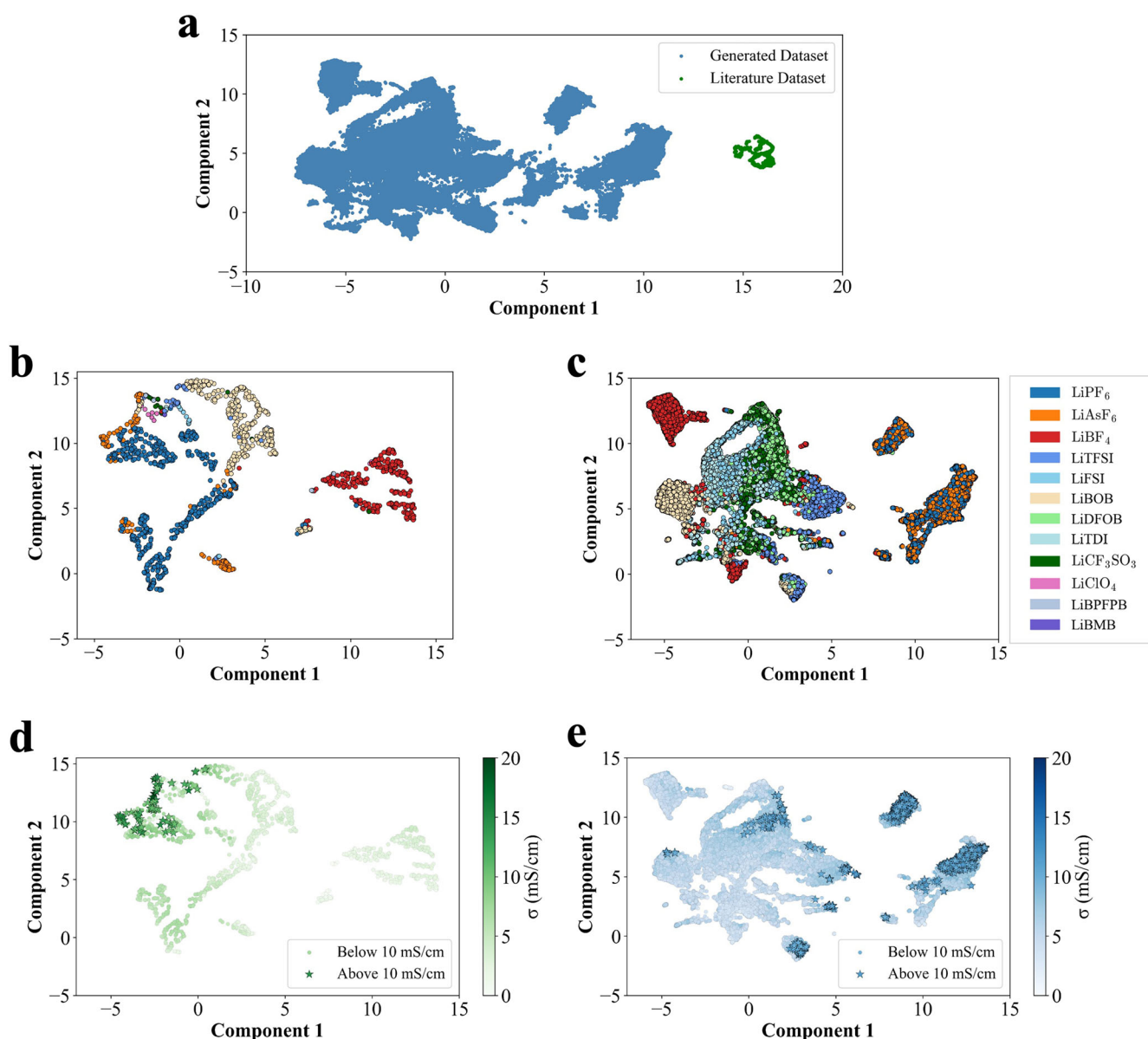


Fig. 4 | Visualization of the SMI-TED-IC formulations embeddings in the generated and literature datasets with UMAP. a UMAP projections of the SMI-TED-IC embedding space to show electrolyte formulation distribution. Datapoints are colored by the source dataset. Individual UMAP projections are colored by the salt

present in the electrolyte for **b** the literature dataset and **c** the generated dataset. The UMAP projections are colored by **d** experimental ionic conductivity in the literature dataset and **e** predicted ionic conductivity in the generated dataset. High ionic conductivity formulations are indicated with star-shaped markers.

a GCN pre-trained on the molecular properties of only ~500 lithium-ion battery-relevant molecules^{27,28}. Thus, the carryover of the supervised pre-training used in the F-GCN model to target electrolyte properties will be contingent upon the relationship between pre-training labels and target formulation properties. Furthermore, the size of labeled datasets will usually be orders of magnitude smaller than unlabeled molecular databases due to the cost associated with calculating labels (properties). This constraint makes F-GCN less transferable to new chemistries or applications where labeled molecular data is scarce, such as in non-Li systems or emerging materials for fuel cells and solid-state electrolytes. Comprehensive mapping of the chemical space enables the SMI-TED-IC model to extrapolate with as few as one example in the training dataset. This demonstrates the unique capability of chemical foundation model-based approaches in exploring the edges of the design space. As demonstrated in this study, such approaches may reduce the dependence on expensive domain-specific simulation datasets, enabling broader, faster, and more cost-effective materials discovery.

A comparison of the literature on machine learning models for liquid electrolyte ionic conductivity prediction is shown in Fig. 3e^{25,26,30,45}. The number of unique molecules is used as an indicator of chemical diversity and represented by the diameter of the marker. The studies that use models such as Gaussian process regression and linear regression with only concentration as input features (dark blue and green markers) are highly accurate, but are constrained to a small number of chemical constituents, and therefore unable to extrapolate to new molecules^{30,45}. The work from Zhu and colleagues (purple marker) utilizes a graph and physics-based model that uses chemical representations as input, but they focused on optimizing a fixed, five-component system. Another graph-based model is applied in the study by Zhang and colleagues to explore a diverse chemical space that included both polymeric and small-molecule containing electrolytes²⁵. However, the accuracy across this broader design space is lower (pink marker) than the other models, which can be attributed to the large diversity and small dataset size. By contrast, the F-GCN model used in the present work achieves the second-best accuracy (yellow marker) over a

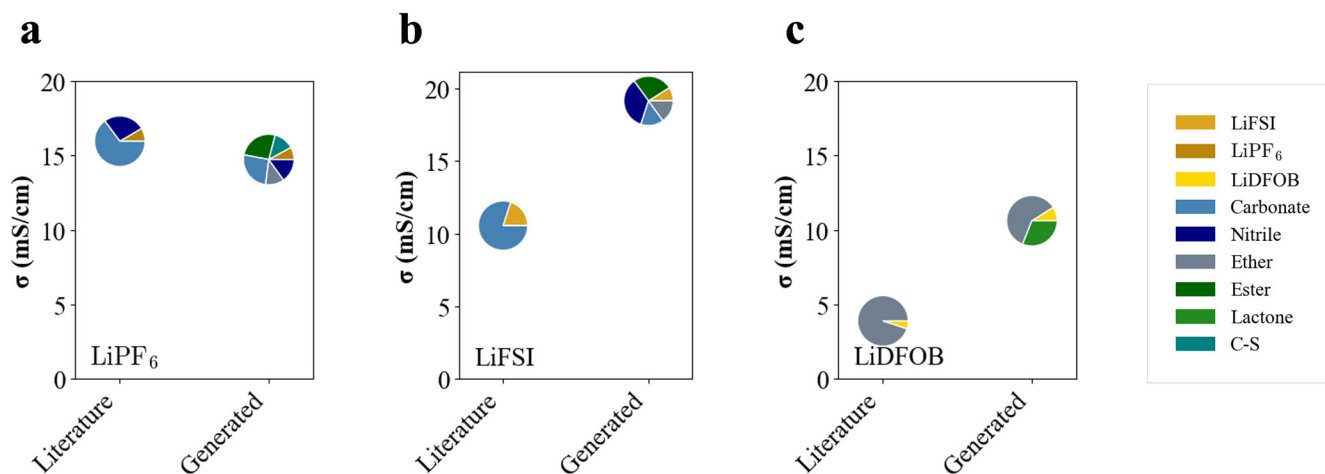


Fig. 5 | Validated novel electrolyte formulation designs for three lithium salts. Comparison of the most conductive electrolytes from the literature and validated electrolytes from the screened list based on the following salts: **a** LiPF₆, **b** LiFSI, and

c LiDFOB. The concentration of the salt and solvent classes present in each electrolyte is indicated in the inset pie charts.

diverse, large dataset, only second to the SMI-TED-IC due to the limitations in extrapolation as discussed above. The work presented here is the first to demonstrate prediction accuracy on par with the state-of-the-art models in the literature over a large and chemically diverse dataset. For this reason, the SMI-TED-IC model is next used to screen novel electrolyte designs to identify high-performing formulations.

Design of novel high-conductivity electrolyte

The demonstrated generalizability of the SMI-TED-IC model leads us to probe unexplored areas of the electrolyte design space to discover novel high-conductivity formulations. To that end, a “generative screening” campaign is performed by first computationally generating 10⁵ electrolyte formulations that are then screened by the SMI-TED-IC model. To implement screening, a threshold must be defined for identifying novel high-conductivity electrolytes. It must be noted that the desired ionic conductivity for a lithium-based battery will depend on multiple factors such as cell chemistry, electrode design and applications. Prior electrolyte discovery efforts have chosen various ionic conductivity thresholds for screening novel electrolytes^{24,59}. Here, we choose a threshold of 10 mS/cm at room temperature (20 °C) based on the conductivity of 1 M LiPF₆ in EC/DMC/EMC (1:1:1 v/v/v)^{60,61}, an electrolyte formulation currently used for evaluating lithium-ion batteries.

The generated dataset samples the immense combinatorial design space of single salt and solvent combinations from the shortlisted set of molecules available for validation. All molecules in the formulation generation chemical pool (listed in Table S2) were sourced from the curated literature, ensuring that each component has prior relevance to rechargeable lithium-ion battery research. Unconstrained generation of formulations based on the chemical pool could lead to a combinatorial explosion of the generated design space to more than a billion samples, making it intractable to screen relevant formulation designs for the study. To avoid this, we applied several constraints in the synthetic generation of electrolyte that are further detailed in the “Methods” section. Generated electrolytes contain up to five solvents from diverse classes, resulting in new and more complex electrolyte formulations compared to the literature dataset. The formulations from the generated dataset and literature dataset (room temperature) are visualized through UMAP projections of the SMI-TED-IC formulation embeddings (Fig. 4a). Here, the distance between embeddings reflects the relationship between molecular structure and ionic conductivity learned during the fine-tuning process. The distance of the literature electrolyte embeddings (green) from the generated dataset (blue) in the UMAP plot indicates that the new salt and solvent combinations in the generated dataset

are outside the training set distribution. To investigate the finer features of formulations in each cluster, UMAP projections of the individual datasets are created and compared. The electrolytes in both UMAP plots form distinct clusters based on the constituent lithium salt (Fig. 4b, c). This distribution is expected given the single-salt constraint in the formulation design, and because the structural differences between the lithium salts will have a greater impact on the transport properties of electrolytes than the relatively contiguous structural differences between the organic solvents used in lithium-ion batteries^{52,62,63}. Figure 4d, e compares the literature-derived and model-learned trends in electrolyte ionic conductivity among the clusters. This distribution of ionic conductivity can also be used to evaluate the significance of the salt domain on performance. In Fig. 4d, the high ionic conductivity formulations (~96 electrolytes) are indicated with star-shaped markers in the top left, region associated with the clusters of LiPF₆, LiAsF₆ and LiBOB containing electrolytes. Meanwhile, the rightmost clusters of LiBF₄ containing electrolytes have low ionic conductivity. This mapping is consistent with the solvation and ionic mobility characteristics of these salts in organic solvents⁵² and with the range of ionic conductivities for each salt in the literature dataset (Fig. S3). The SMI-TED-IC identified 3351 high ionic conductivity formulations within the generated dataset (indicated by a star-shaped marker in Fig. 4e) that are dispersed more widely across salt domains, suggesting the potential for high-conductivity formulation designs with new salts.

To validate the model’s ability to identify electrolyte designs with high ionic conductivity, we shortlist 16 electrolyte formulations from the screened group for experimental testing based on the highest predicted conductivity within multiple salt domains. Ionic conductivity measurements are made experimentally following the procedure described in the supplementary information (SI). Several generated electrolyte formulations demonstrate high ionic conductivities as listed in Table S3. Figure 5 compares the solvent formulation design and experimental ionic conductivity of the best electrolytes in the literature dataset and the new electrolytes (generated and validated) for three salts. The new electrolyte formulation containing LiPF₆ is shown in Fig. 5a (generated) and has an ionic conductivity on par with the highest conductivity LiPF₆ electrolyte reported in the literature⁴⁶. However, the pie-chart in the figure inset indicates an entirely new solvent formulation design with five different solvent classes to achieve high ionic conductivity. This new design is unique from the literature dataset, where electrolytes constitute at most three solvent classes. This formulation also constitutes an unusual component, dimethyl sulfoxide (DMSO), a linear organosulfur (C-S), which is not part of any high-conductivity formulations in the literature dataset (Fig. S4). Since the

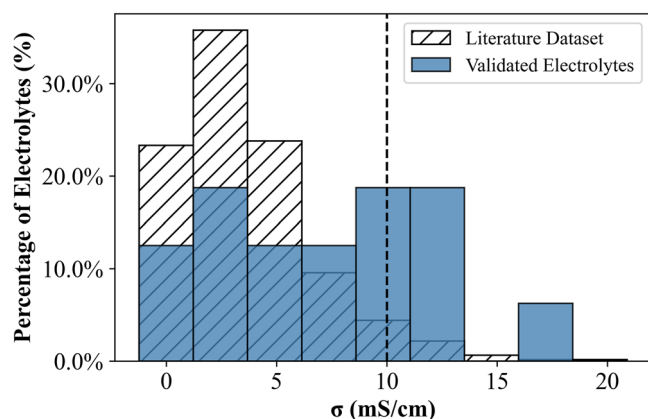


Fig. 6 | Ionic conductivity distribution in the literature dataset compared to foundation model-guided designs. Histogram of experimental ionic conductivity measurements at room temperature from the literature dataset and from novel electrolytes screened by the model and experimentally validated.

emergent properties of multicomponent electrolyte mixtures have been shown to improve ionic conductivity^{50,64}, the discovery of this novel complex formulation emphasizes the scope of chemical foundation models in material discovery, especially within multicomponent materials due to the exponential growth of the potential design space with each added component⁷.

The ionic conductivity of the new LiFSI electrolyte (generated) is 19.14 mS/cm, which is 82% higher than the best in literature (Fig. 5b). While, both the literature and generated electrolytes contain dimethyl carbonate, the generated electrolyte additionally include an ester (methyl formate), a nitrile (propionitrile), and a cyclic ether (2-methyltetrahydrofuran), resulting in a compositionally distinct solvent system. To our knowledge, there has not been any prior report on a methyl formate-containing electrolyte with an ionic conductivity near 19.14 mS/cm. Our literature review suggests that methyl formate specifically has been reported in a small number of studies focusing on lithium-ion batteries for low temperature applications^{65,66}. Its inclusion in this high-performing candidate is therefore chemically plausible and suggests potential for broader applicability beyond niche temperature ranges. We further note that the model prediction for the conductivity of this electrolyte is 10.52 mS/cm, an underprediction that represents the extrapolative challenge of modeling electrolyte systems that are different not only in chemistry (new solvent classes) but also in formulation complexity (a greater number of components). Although the foundation model showed improved generalization for sparsely represented molecules in the literature dataset (Fig. 3c, d), the generated electrolytes lie in a region of the design space that is both chemically and combinatorially underexplored. We remark that addressing this challenge may require mitigation strategies such as uncertainty-aware predictions⁶⁷, active learning⁶⁸, and the development of more chemically diverse training datasets with high-throughput experiments^{7,31,45} as part of future work. Despite these limitations, the present model successfully identifies promising trends and compositional motifs, supporting its utility in guiding formulation discovery.

Another instance of significant conductivity improvement in a discovered electrolyte is observed with LiDFOB, where the generated electrolyte is 172% more conductive than the literature counterexample (Fig. 5c). Interestingly, there are only two formulations with LiDFOB in the training data, both of which have dimethoxyethane (DME) as a sole solvent. While LiDFOB has benefits in SEI formation and thermal stability, the ionic conductivity of LiDFOB electrolytes with ethers and carbonates is lower than the industry standard LiPF₆-containing electrolytes⁶⁹. With our model-guided approach, we discover an electrolyte with γ -butyrolactone (GBL), diglyme, and 2-methyltetrahydrofuran that improves the conductivity above most electrolytes in the dataset. The synergy between GBL and

LiDFOB particularly has led to renewed interest in its use as the primary salt, and recent studies of this system demonstrate promising low-temperature battery cycling performance due to high conductivity across a wide range of temperatures^{19,70,71}. Recognizing an emerging area of electrolyte design without prior training is a key finding that demonstrates the extrapolation capabilities of this model.

Discussion

Properties of several materials, such as composites and polymer blends, are governed by their compositions^{59,72}. Accordingly, numerical representations of materials that map their structure and composition to a property have been of significant interest. In the case of battery electrolytes, compositions play a pivotal role in determining the overall performance of the collective mixture. Thus, the incorporation of compositional information in a formulation model is essential. Our approach presents an opportunity to leverage rich chemical embeddings from a pre-trained foundation model and combine them with compositions to create a complete representation of multi-constituent materials such as mixtures and formulations. While different modes of aggregation of chemical and composition embeddings can be performed as elaborated in our prior works^{42,43}, the present framework of formulation representation as a string enables efficient fine-tuning of the language model on the formulation designs, delivering improved generalization. Furthermore, a simplified string representation of formulations/mixtures opens the future scope of applying inverse design strategies for the discovery of such materials.

One challenge with respect to electrolyte discovery is that quantitative design rules for formulations are often elusive due to the complex, nonlinear interactions between constituents. The optimal concentration of a given component is not an isolated parameter but is instead dependent on its interaction with other species in the mixture. This multivariate interdependence, combined with the vast number of possible combinations, renders comprehensive, rule-based mapping of formulation criteria practically infeasible^{73,74}. Further complicating interpretation is the growing use of nonlinear machine learning models, such as the transformer-based SMI-TED, which rely on tokenized molecular inputs. These models, while powerful in predictive capability, present challenges for interpretability due to their intrinsic abstraction of chemical features into learned embeddings. For this reason, many studies focus on identifying suitable molecular designs for individual constituents, rather than attempting to rationalize or optimize full multicomponent formulations^{25,75}. Our effort to quantify the relationship between the concentration of solvent classes and ionic conductivity in the generated dataset (predicted ionic conductivity) is shown in Fig. S5 using the Spearman correlation coefficient. Overall, the resultant correlations within the formulation design are mostly weak and non-informative, underscoring the limitations of pairwise analysis in such complex systems, and highlighting the importance of developing interpretability methods suited for multicomponent, nonlinear design spaces.

Despite these challenges, the model-guided exploration yielded electrolyte candidates with predicted ionic conductivities that exceed those of most entries in the literature-derived training set. This suggests that the discovered formulations occupy underexplored regions of the design space that nonetheless adhere to constraints informed by prior knowledge. By identifying formulations that combine known components in compositions not previously emphasized, the model offers a means to prioritize candidates that may perform beyond expectations based on individual component properties alone. These results reinforce the potential of machine learning frameworks to guide formulation design by uncovering promising combinations that merit further investigation, both experimentally and mechanistically.

Comparison of the ionic conductivity distribution of all validated electrolytes with the literature-extracted dataset in Fig. 6 attests to the likelihood of finding high-performing electrolyte formulations with the present approach. 7 of the 16 validated electrolytes had a conductivity above the threshold value of 10 mS/cm, implying a 44% probability of discovering a new electrolyte design with targeted high performance based on the

present approach. The success rate in the inverse design of materials and molecules reported in the literature is highly variable, given differences in the materials system, outcome metric, and validation protocols. However, the majority of these studies use first principles calculations to validate the thermodynamic stability of the generated structures rather than experimental validation, reporting a ~10% success rate⁷⁶. While the success rate of 44% from the generative screening approach is promising, it can be further improved by using metaheuristic optimization methods such as genetic algorithms to guide materials generation, as shown in a recent study on self-assembling peptide discovery⁷⁷. Such approaches remain to be realized and evaluated for formulation design in future work. We also assessed the ionic conductivity predictions on generated electrolytes using F-GCN in Fig. S6. The results highlight the tendency of the F-GCN model to overpredict ionic conductivity at higher values, reducing the probability of finding high-performing candidates in the screening process. A comparison of model predictions on each validated electrolyte is shown in Table S4. Though F-GCN is a powerful approach, SMI-TED-IC presents advantages in finding trends in out-of-distribution electrolytes while bounding the design space to facilitate novel electrolyte discovery.

It should also be noted that ionic conductivity is only one of several factors to consider when selecting or optimizing an electrolyte for a given battery application. Practical deployment also requires consideration of electrochemical stability (to ensure compatibility with the voltage window of the electrodes), thermal and chemical stability, formulation solubility, viscosity, and safety-related factors such as flammability or reactivity with moisture. Therefore, to tailor the current high-conductivity screening results to specific battery systems, additional down-selection stages may be necessary, incorporating these complementary criteria to identify formulations that meet the full set of operational and safety requirements. While a comprehensive evaluation of all relevant properties is beyond the scope of this study, future work may combine models for multiple electrolyte properties to enable multi-objective screening and accelerate the discovery of application-ready formulations.

In summary, integration of a chemical foundation model in the material discovery workflow makes the approach generalizable across a combinatorial chemical space, as shown in the present work. The model maintains good prediction accuracy for molecules that are sparsely covered in the training data, like fluorinated ethers and the LiNO₃ salt. Furthermore, the model demonstrates a consistent ability to identify high-performing formulations across multiple electrolyte chemistries, as validated for three different lithium salts. Lastly, the scope of the SMI-TED-IC model in discovering novel and high-performing formulations is highlighted in the discovery of newly reported LiDFOB and LiFSI electrolyte formulations with 172% and 82% improvement in conductivity, respectively, despite training data deficiency. By adopting a pre-trained foundation model for chemical representation, we reduce the dependence on the quantity of labeled formulation datasets, which could be laborious and challenging to acquire in several cases^{27,28,34}. Overall, this work showcases the power of foundation models to unlock the full potential of data by bridging knowledge across complex scientific design spaces, paving the way for practical machine learning tools that can actively guide and accelerate experimental discovery.

Methods

Data curation and processing

Experimental ionic conductivity values of liquid electrolytes for non-aqueous lithium-based batteries are collected from the lithium-ion battery electrolyte literature (Table S1). The criteria for selecting journal articles include dataset size and chemical diversity. The largest source of data is the CALiSol-23 dataset, which combines electrolyte formulations and ionic conductivity values from 27 journal articles⁵¹. The electrolyte dataset includes the electrolyte constituents, their respective concentrations, the experimentally measured ionic conductivity, and the temperature at which the measurement is taken. Each molecule in the dataset is represented by the simplified molecular-input line-entry system (SMILES) that is converted to

canonical SMILES using the RDKit package to ensure that each molecule has a unique SMILES string. The concentration of each component is converted from the reported unit to molar percentage (mol%) by using the molecular weight and density of the chemical constituent. Additionally, the conductivity values are transformed to a logarithmic scale because ionic conductivity spans multiple orders of magnitude (Fig. S7). The temperature is converted from the reported units to Celsius, and the values range from 200 to −70 °C (Fig. S8). To mitigate the impact of experimental noise, we curated the dataset by prioritizing studies with replicated measurements and standardized calibration procedures. Additionally, outlier datapoints were systematically identified through visual analysis of conductivity distributions using box plots and removed to reduce the influence of anomalous values on model training. In the case of a duplicate electrolyte formulation at the same temperature, the ionic conductivity value is averaged.

SMI-TED-IC model development

We employed the SMI-TED a large-scale, open-source model, pre-trained on a curated dataset of 91 million SMILES samples from PubChem, comprising 4 billion molecular tokens, and having 289 million parameters⁴⁴. SMI-TED incorporates an encoder-decoder architecture that significantly enhances model performance and stability over previous foundation models that are based on an encoder-only transformer model and lacked a natural inversion process⁴². The added decoder function in the SMI-TED model overcomes this issue by allowing better representations and denoising. The pre-trained model is customized and fine-tuned for downstream specialized tasks such as predicting an electrolyte formulation property.

A dataset of 13,666 electrolyte formulations and their reported ionic conductivities at different temperatures is curated from the literature. The dataset consists of formulations containing up to six electrolyte constituents, where each constituent is represented by its canonical SMILES string. The concentration of each constituent molecule is expressed as the fraction of its molar% in the formulation. The complete dataset is divided into train, validation, and test subsets by random splitting into 80-10-10 ratios. Next, to generate a good feature representation of electrolyte mixtures, featurization steps are applied sequentially to the train and validation datasets. First, a concatenated sequence (*seq_A*) of SMILES of formulation constituents and their respective concentrations is generated, where each component is unified by the <sep> token as indicated in Fig. 1b. Concentration values are inserted into the input string as numeric tokens and embedded alongside SMILES tokens using the same tokenizer and embedding layer. This allows the model to learn joint attention over both molecular identity and relative abundance. While this approach may limit the model's sensitivity to small numerical variations in concentration, it provides robustness against overfitting and supports effective pattern recognition across a large and diverse formulation space⁷⁸. As formulant count varies across the whole dataset, formulations containing less than six constituents are padded with dummy features such as SMILES of water ('O'), with their respective composition designated as 0.0%. The use of a fixed-length input with padding ensures consistency across varying mixture sizes. Next, the dataset is augmented to include all possible permutations of the constituent SMILES order, followed by their respective compositions in the formulations to ensure the model is invariant to the order of formulants in the representation (Table S5). This step ensures that the model does not learn any unnecessary patterns, such as the sequence of formulation constituents in the dataset, especially since the algorithm is based on a transformer architecture where the learning is sensitive to the order of tokens in the inputs. It is important to highlight that the test dataset is not augmented to preserve fair evaluation of the model. The resultant feature strings (*seq_A*) are then transformed into chemical embedding vectors by a pre-trained SMI-TED model (*seq_C*) and then aggregated along with their corresponding temperatures at which ionic conductivity has been recorded (*seq_F*). Prior to concatenation, temperature is standardized to match the scale of hidden features, mitigating concerns about numerical imbalance. This approach allows the model to learn an appropriate weighting of the temperature variable without increasing model complexity. Thus, the final vector (*seq_F*)

captures complete chemical, compositional, and conditional (temperature) information of electrolyte formulations and is used as an input feature in the fine-tuning process.

The feature vectors (seq_F) obtained after the featurization step are used as the inputs in the fine-tuning of the SMI-TED model for the downstream task of predicting ionic conductivity. The pre-trained SMI-TED289M model is fine-tuned to learn electrolyte representations (seq_F) and predict their conductivities. The fine-tuning is conducted over 1200 epochs on the augmented training dataset using a fixed learning rate of $3e-5$ and a batch size of 128 seq_F . During fine-tuning, loss on train and validation datasets is calculated as mean absolute errors (MAEs) and prediction on test dataset is evaluated in root mean squared error (RMSE). A schematic of the fine-tuning process is shown in Fig. 1b.

Electrolyte candidate generation and screening

To demonstrate the ability of this approach to discover novel electrolyte formulations, we generate 100,000 dummy electrolyte formulations from combinations of the solvents and salts present in the literature-extracted dataset and use the SMI-TED-IC model to screen for high ionic conductivity formulations. Certain electrolyte designs would be nonviable as electrolytes, such as those consisting of either only solvents or only salts. Also, electrolytes with high salt concentrations (>15 mol%) are screened out to avoid solubility issues. Therefore, generated electrolytes constituting a single salt with a concentration between 5 and 15 mol% are further considered. The number of salts is constrained to one to match the literature dataset, while the solvent concentration range is constrained to above 10%. The molecule pool is filtered for lab availability, and components and concentrations are randomly sampled to generate dummy electrolyte formulations with two to six components. The ionic conductivity of each formulation is predicted at room temperature (20 °C) using the SMI-TED-IC model, and a subset of these electrolytes is selected for experimental validation based on ionic conductivity and salt diversity. We highlight that these generation rules are subject to domain expert, application and chemical availability.

Ionic conductivity measurements. To validate electrolytes screened by the model, ionic conductivity measurements are performed in symmetric coin-cells with 15.5 mm diameter stainless steel blocking electrodes using electrochemical impedance spectroscopy (EIS) on a multi-channel potentiostat (Biologic, VSP). A quartz microfiber separator soaked with 100 μ L of electrolyte is sandwiched between the blocking electrodes. All cells are fabricated in an Ar-filled glovebox (<1 ppm O_2 , <1 ppm H_2O). The frequency range for the EIS measurement is 100 Hz to 1 MHz, and a 5 mV AC voltage amplitude is applied. The DearEIS Python module is used to fit the EIS data with an equivalent circuit. The equivalent circuit consisted of an ohmic resistor (R_1) in series to a constant phase element (CPE1) and another ohmic resistor (R_2) which are in parallel. The value of R_1 is used as the solution resistance. During experimental validation, each formulation is measured in triplicate, and the average value is reported to reduce random measurement errors. All results are calibrated against a 1 M LiPF₆ in EC/DMC (1:1 v/v) standard to correct for systematic variations across measurements. A cell constant is calculated based on the dimensions of the cell and validated experimentally with standard carbonate-based lithium-ion battery electrolyte solutions. The cell constant is multiplied by the solution resistance to get the ionic conductivity value of the electrolyte. All experiments are conducted at room temperature (20 °C).

F-GCN model. A modified formulation graph convolutional network (F-GCN) is used to predict ionic conductivity from electrolyte formulation and temperature. Each constituent is represented by its SMILES string, converted to a molecular graph using RDKit, and encoded into a 100-dimensional Graph Representation (GR) vector using a pre-trained GCN. These GR vectors, scaled by the molar fraction of each component, are concatenated along with the temperature to form the model input. This composite descriptor is passed through a feed-forward neural

network to predict ionic conductivity. Hyper-parameter tuning is performed to customize the previously published F-GCN model for ionic conductivity prediction^{27,28}. The GCN model for the molecular representation is not altered, but the aggregator function leading to formulation descriptor and feed-forward model hyper-parameters are systematically tuned and evaluated for the lowest error. The best performing model used concatenation as the aggregator function for the formulation descriptors and a feed-forward neural network with two hidden layers that consist of 200 and 100 nodes, respectively.

Data availability

Ionic conductivity datasets associated with the study can be found in the GitHub repository: https://github.com/murtazazohair/IBM_SMI-TED-IC.

Code availability

The code for ionic conductivity prediction with the SMI-TED-IC model associated with this study can be found in the GitHub repository: https://github.com/murtazazohair/IBM_SMI-TED-IC.

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Author contributions

M.Z., V.S., E.V.B. and Y.H.L. conceptualized the project. M.Z. and E.A.S. developed and trained machine learning models. M.Z. curated literature datasets. M.Z., V.S., and M.G. contributed to the analysis. K.N., L.S., and A.T. contributed to the experimental validation of ionic conductivity. M.Z. and V.S. wrote the original draft. All authors contributed to draft writing and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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